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Improving clustering stability in intelligent vehicular networks using Graph Neural Networks

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Abstract

The rapid evolution of wireless technologies and the automotive industry have positioned Intelligent Vehicular Networks (IVNs) as a fundamental pillar of Intelligent Transportation Systems (ITS). This paper presents a comprehensive methodological framework for IVNs clustering using advanced Graph Neural Network (GNN) architectures. We address critical challenges such as cluster stability, connectivity duration, and accuracy in highly dynamic highway environments. Our improved GNN model surpasses traditional clustering methods, including k-means, spectral clustering, and graph auto encoders, achieving 99.4% accuracy while maintaining an average cluster connectivity duration of 890 s (nearly 15 min). This performance was observed in a realistic simulation environment modeling a 5 km highway segment with 3 lanes, a vehicle density of 150, and a communication range of 200 m, reflecting dynamic traffic conditions. The proposed approach integrates realistic mobility models based on PeMS database characteristics and introduces new evaluation metrics for temporal cluster stability analysis. Experimental results demonstrate significant improvements in cluster lifetime, stability, and connectivity compared to state-of-the-art techniques.

Keywords IVNs, Graph Neural Networks (GNN), Clustering, Cooperative driving, Intelligent Transportation Systems (ITS)

1 Introduction

With the rapid increase in road traffic, Intelligent Vehicular Networks (IVNs) have become a cornerstone of Intelligent Transportation Systems (ITS) [1]. These networks facilitate decentralized and continuous communication, not only between vehicles (V2V) but also with roadside infrastructure (V2I). This interaction generates massive data flows essential for developing cooperative driving and traffic optimization applications [2]. A major challenge in this context is ensuring the connectivity of vehicle groups, or *clusters*, over extended periods. Stable connectivity is indeed crucial for guaranteeing the reliability and efficiency of information exchange, which forms the basis of safety and performance in ITS.



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Managing IVNs, particularly in highly dynamic environments such as highways, presents significant complexity. The primary difficulty lies in maintaining cluster cohesion for durations sufficient to enable reliable communication. Rapid topological changes, caused by high vehicle mobility, lead to frequent link disruptions, which significantly degrade the performance of cooperative applications [3]. Unlike traditional approaches that focus mainly on selecting *cluster heads*, our analysis emphasizes improving the overall connectivity and temporal stability of the entire cluster. Conventional clustering methods struggle to adapt to this dynamic nature, resulting in limited cluster lifetimes and network fragmentation (Fig. 1).

1.1 Contribution: A GNN-based approach for enhanced stability

Traditional clustering methods fail in dynamic highway scenarios, leading to short-lived clusters. Graph Neural Networks (GNNs) offer a powerful solution. By treating the vehicle network as a graph, GNNs improve cluster stability and lifetime [4].

This paper presents a novel GNN-based model for VANET clustering that outperforms traditional techniques like k-means and spectral clustering. Our approach uses realistic mobility data from the PeMS database [5] and introduces new metrics to measure temporal stability, providing a robust method for evaluating performance in real-world conditions.

Our main contributions are as follows:

1. **A Novel GNN-Based Clustering Model:** We introduce advanced Graph Neural Network (GNN) architecture specifically designed for clustering in highly dynamic IVNs. This model is built to capture complex inter-vehicle relationships to enhance cluster stability over time.
2. **Superior Performance Over Traditional Methods:** We provide a comparative evaluation demonstrating that our GNN-based approach significantly extends cluster lifetime compared to conventional techniques, including k-means clustering, spectral clustering, and graph autoencoders.
3. **Realistic Evaluation Framework and New Metrics:** We validate our model using realistic mobility scenarios derived from the PeMS database. Furthermore, we propose and integrate novel metrics specifically designed to quantitatively assess the temporal

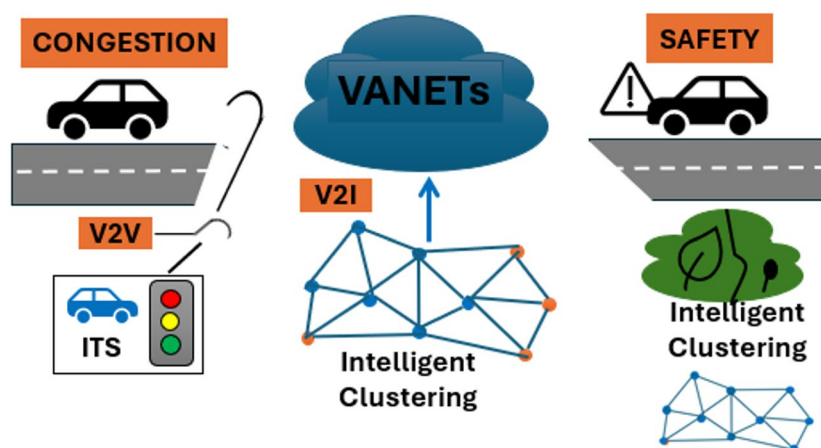


Fig. 1 Architecture and Role of IVNs in ITS

stability of the resulting clusters, offering a more accurate measure of performance in dynamic environments.

1.2 Paper organization

To present our contribution, the paper is structured as follows: Sect. 2 presents the state of the art in clustering methods. Section 3 provides a detailed description of our GNN-based clustering approach along with the proposed architecture. The following sections are dedicated to experimental evaluation, discussion of the results, and finally, the conclusion and future work.

2 Related works

Clustering in vehicular networks is an essential technique for improving routing scalability, network robustness, and the stability of cooperative environments. By grouping vehicles according to correlated spatial and dynamic criteria, these clusters form the foundation of critical ITS applications such as accident detection, congestion management, and information dissemination.

2.1 Traditional and distributed approaches

Early clustering methods in IVNs are largely inspired by techniques developed for Mobile Ad Hoc Networks (MANETs) [4]. Most adopt a distributed approach, relying on Dedicated Short-Range Communications (DSRC) without the need for centralized infrastructure. These algorithms primarily focus on the optimal selection of cluster heads (CH), as group stability largely depends on this choice [3]. Weighted criteria used in CH selection include factors such as position, speed, and direction [6]. However, these traditional methods often generate substantial communication overhead and struggle to maintain efficiency in highly dense and highly dynamic environments [7] (Fig. 2).

2.2 Machine learning-based clustering

To overcome the computational complexities of traditional distributed methods, machine learning has been integrated into clustering algorithms for IVNs. The k-means algorithm and its improved variants remain popular choices due to their simplicity and

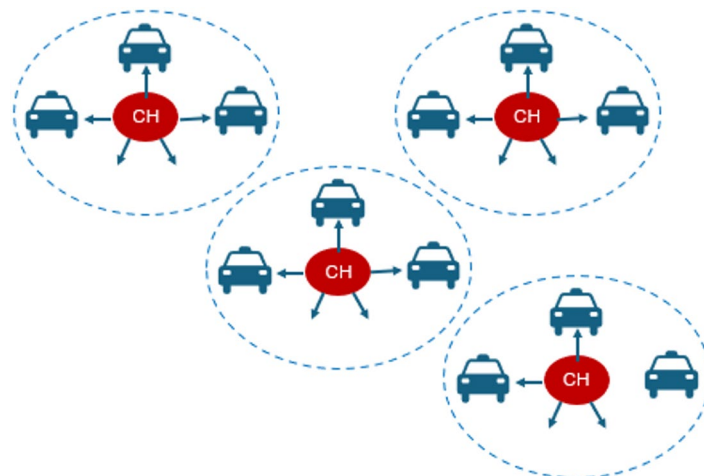


Fig. 2 Distributed clustering architecture based on Cluster Head selection

efficiency [8]. Some studies combine these techniques with fuzzy logic to predict and stabilize cluster formation by considering future vehicle movements. Other advanced solutions use spectral clustering, based on spectral decomposition of graph matrices, to enhance group stability in dynamic contexts. Despite their contributions, these methods still face limitations, particularly in fully capturing the topological and temporal complexity of vehicular networks [9] (Fig. 3).

2.3 Graph Neural Networks (GNN) in ITS

GNN-based clustering approaches have opened new perspectives for processing data in non-Euclidean domains, such as complex graphs representing IVNs [10]. These architectures learn to encode vehicle relationships and features into optimized vector representations, enabling the formation of coherent clusters.

Several recent studies have demonstrated the effectiveness of GNNs in modeling Intelligent Transportation Systems (ITS). For example, GNNs have been widely applied in traffic prediction, anomaly detection, and modeling of complex vehicle interactions [11].

Regarding clustering, GNNs move beyond traditional cluster-head selection and instead focus on improving the overall clustering quality by maximizing group connectivity and cohesion. Previous works have explored the use of GNNs to enhance cluster stability in IVNs. Our approach follows this innovative paradigm by reformulating clustering as a vehicle aggregation task, grouping vehicles with similar behaviors into the same cluster thus ensuring longer connectivity and improved network stability.

Our proposal aligns with this evolution by implementing an optimized GNN model capable of handling dynamic inter-vehicle relationships through multi-layer architecture and a multi-restart k-means clustering procedure to ensure robustness and stability (Fig. 4).

Table 1 presents a detailed comparison of different clustering methods proposed in recent VANET literature.

Table 1 presents a synthesis of the most relevant and recent approaches, covering a diversity of methodologies ranging from models based on Graph Neural Networks

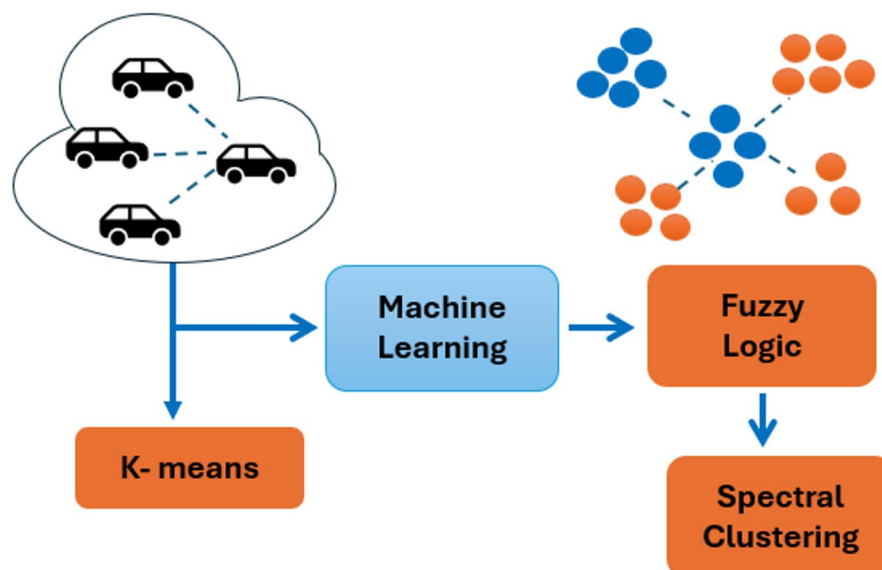


Fig. 3 Machine Learning-Enhanced Clustering Techniques for IVNs

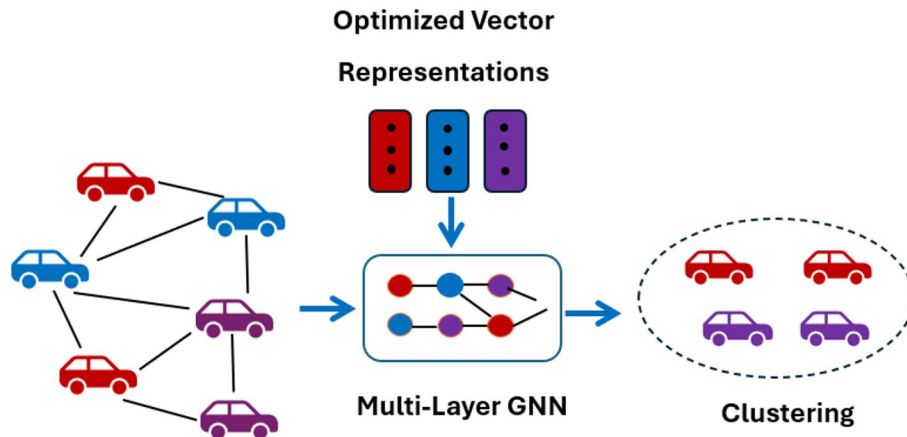


Fig. 4 GNN-Based Clustering Architecture for IVNs

(GNN) and Deep Reinforcement Learning (DRL) to nature-inspired techniques and analytical models. The analysis of this state-of-the-art reveals a marked trend towards the use of machine learning to improve cluster stability and security, while also highlighting persistent challenges. Existing works show notable shortcomings regarding sensitivity to structural noise, vulnerability to adversarial attacks (such as graph perturbation, policy manipulation, or Centroid Shifting), generalization complexity, and memory and computational resource cost. Our work is precisely positioned to address these gaps by proposing an improved GNN proposed approach which, while leveraging the power of graph architectures, integrates realistic mobility models and temporal evaluation metrics to ensure unparalleled cluster stability and connectivity duration in highly dynamic highway environments.

3 Proposed approach: GNN-based methodology

This section presents the detailed proposed approach for stable and efficient clustering in VANETs and IVNs, which is centered around a novel Graph Neural Network (GNN) framework. The methodology is structured into four main components: first, the Construction of the VANET Graph which models the dynamic connectivity between vehicles; second, the Data Preparation which defines the feature vectors for the GNN input; third, the GNN Model Architecture which is designed for robust feature aggregation; and finally, the Clustering Algorithm which leverages the learned representations for superior performance compared to traditional methods.

3.1 Construction of the IVN graph

In our method, each vehicle is represented by a node $v \in V$ within a dynamic graph.

$G_t = (V, E_t)$ constructed at each time instant t . The edges $e \in E_t$ are established between nodes based on immediate spatial proximity, with a threshold set at approximately 200 m, corresponding to the typical communication range in a vehicular environment.

Each node is associated with a feature vector X_v that captures the essential attributes of the vehicle:

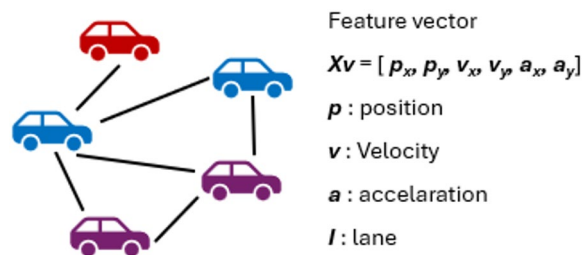
$$X_v = [px, py, vx, vy, ax, ay, l]$$

Table 1 Summary of previous studies

Ref	Year	Objectives	Classifier	Adversarial Technique	Dataset	Limitations
Hu & Lee – GNN-based Clustering Enhancement [12]	2022	Improve cooperative clustering in VANET	GNN	FGSM Graph Perturbation	VANET Simulations	Sensitive to structural noise
Marwah & Jain – Hybrid Optimization + Ensemble Learning [13]	2022	Improve network stability using hybrid optimization	Ensemble Learning	Data Poisoning Attack	Simulation	High vulnerability to poisoned samples
Talib et al. – Center-Based Stable Clustering [5]	2015	Realistic Mobility and Channel Modeling	Analytical Models	Analytical Models	Real Mobility Data	Complexity, Limited Generalizability
Graham et al. – Edge Clustering with GNN [15]	2023	Enhance edge communication efficiency	GNN	PGD Attack on Graphs	IoV Simulation	Scalability issues
Gayathri – CSKAS-VANET [16]	2024	Provide secure and stable clustering	RBMA + Signcryption	Spoofing Attack	Simulation	High crypto overhead
Yerrathi et al. – Nature-Inspired Stability [17]	2024	Improve VANET stability using bio-inspired algorithms	Nature-Inspired Model	Jamming Attack	Simulation	Affected by channel interference
ArXiv – GNN + DRL Dynamic Clustering [18]	2024	Enable dynamic clustering with DRL + GNN	DRL + GNN	Adversarial Policy Perturbation	Simulation	DRL policies easily manipulated
Shandilya – VAST-GCN [19]	2025	Robust cluster-head selection	Attention-GCN	Graph Injection Attack	Simulation	Expensive memory usage
Connectivity-Based Clustering (MDPI) [20]	2021	Connectivity-driven clustering	Connectivity Clustering	Replay Attack	Simulation	Poor detection of repeated identities
Coalitional Graph Game [21]	2022	Maximize multi-hop clustering coverage	Graph Game Theory	Strategic Manipulation Attack	Simulation	Vulnerable to malicious coalition members
K-Means Powered Algorithm [22]	2023	Improve clustering efficiency	K-Means	Centroid Shifting Attack	Simulation	K-Means easily manipulated
Clustering Approach to Improve VANET [23]	2023	Enhance VANET performance	Classical Clustering	Label-Flipping Attack	Simulation	No robust mechanisms
Edge Clustering (duplicate of #4) [24]	2023	Optimization of edge communication	GNN	Evasion Attack	IoV Simulation	Duplicate study

Table 1 (continued)

Ref	Year	Objectives	Classifier	Adversarial Technique	Dataset	Limitations
GNN for Routing Optimization – Challenges & Opportunities [25]	2024	Survey challenges of GNNs for routing	Various GNN models	FGSM, Nettack, Graph Poisoning	—	Theoretical review
Graph Contrastive Learning for Clustering [26]	2025	Node clustering using contrastive learning	Graph Contrastive Learning	Embedding Attack	Simulated Graph Datasets	High computational cost

**Fig. 5** Construction of the VANET Graph

where p represents the position, v the velocity, a the acceleration, and l the lane being used. This feature matrix X serves as the input to the GNN model, providing a comprehensive and contextual representation of each vehicle within the network.

Figure 5 illustrates the construction process of the VANET graph, where vehicles are modeled as nodes and edges represent communication links based on spatial proximity.

3.2 Data preparation

In this section, we detail the methodology adopted to prepare the data and model the vehicle clustering problem within the VANET network. Each vehicle is represented by a feature vector X_i at time t :

$$x_i(t) = [p_i(t), v_i(t), a_i(t), l_i(t)]^T$$

where $p_i(t)$ is the position, $v_i(t)$ the velocity, $a_i(t)$ the acceleration, and $l_i(t)$ the lane occupied by vehicle i .

The dynamic graph $G_t = (V, E_t)$ is constructed for each time window t , where:

- V is the set of vehicles (nodes).
- E_t is the set of edges determined by an adjacency matrix A_t , where the existence of a link between vehicle i and vehicle j is defined by:

$$A_t(i, j) = \begin{cases} 1, & \text{if } d(p_i(t), p_j(t)) \leq \delta \\ 0, & \text{otherwise} \end{cases}$$

where $d(\cdot, \cdot)$ is the Euclidean distance between the positions of vehicles i, j , and δ is the proximity threshold (communication range, typically 200 m).

This representation allows the exploitation of spatial and dynamic information for efficient clustering via a graph neural network.

3.3 GNN model architecture

The design of our clustering algorithm is based on a graph neural network architecture composed of multiple SAGEConv (GraphSAGE) layers. These layers efficiently aggregate information from each node's neighbors by combining their own attributes with the aggregated information from their neighborhood.

The first phase, Aggregation, consists in gathering the feature vectors of all neighboring nodes $u \in N(v)$ for the target node v , according to:

$$\mathbf{h}_{N(v)}^{(k)} = AGGREGATE^{(k)} \left(\left\{ \mathbf{h}_u^{(k-1)}, \forall u \in N(v) \right\} \right)$$

In this step, $\mathbf{h}_u^{(k-1)}$ denotes the feature embedding of a neighboring node u from the previous layer $k-1$, while $N(v)$ represents the set of neighbors of v . The function $AGGREGATE^{(k)}$ (e.g., mean or max-pooling) computes a summary vector $\mathbf{h}_{N(v)}^{(k)}$ that captures the aggregated information from the neighborhood.

The second phase, Update, integrates the aggregated neighborhood representation with the previous embedding of the node to produce its updated state:

$$\mathbf{h}_v^{(k)} = \sigma \left(W^{(k)} CONCAT \left(\mathbf{h}_v^{(k-1)}, \mathbf{h}_{N(v)}^{(k)} \right) \right)$$

Here, $\mathbf{h}_v^{(k-1)}$ is the feature vector of node v from the preceding layer. The operator **CONCAT** combines the node's prior representation with the aggregated neighborhood vector. The matrix $W^{(k)}$ corresponds to the trainable weights of layer k , and σ is a non-linear activation function (such as ReLU) that generates the updated embedding $\mathbf{h}_v^{(k)}$ [27].

3.3.1 Role of GNN in enhancing clustering stability

The effectiveness of our approach relies on the GNN's ability to model complex and dynamic relationships within VANETs, overcoming the limitations of traditional methods. The significant improvement in clustering stability is intrinsically linked to the following properties of the GNN:

- **Capturing Spatio-Temporal Dependencies:** The GNN, and particularly the SAGEConv operator, is designed to iteratively aggregate information from neighboring nodes. This aggregation allows the model to capture not only individual vehicle characteristics (position, velocity, acceleration, lane) but also how these characteristics are influenced by the local environment and how these interdependencies evolve over time. This ability to model dynamic relationships is crucial for predicting the future cohesion of vehicle groups.
- **Adaptive Aggregation Functions:** Unlike methods that use static aggregation rules, the GNN learns optimized aggregation functions. These functions dynamically weigh the importance of neighbors and their attributes, enabling the model to identify complex patterns that indicate group stability. This means the GNN can adapt to different traffic configurations and varied driving behaviors to form more coherent clusters.

- **Generating Rich Semantic Embeddings:** The GNN layers transform raw vehicle features into high-dimensional vector embeddings. These embeddings encode rich semantics of vehicle behavior and their interactions within the network. Clustering is then performed on these semantic embeddings, making the formed groups intrinsically more informed and robust to topological changes.

By combining these capabilities, the GNN does not merely use the provided features; it interprets and combines them in a way that anticipates and maintains cluster stability, even in highly dynamic VANET environments.

3.4 Clustering algorithm

Our study proposes a clustering algorithm based on GNN, designed to capture the complex relationships between vehicles in a dynamic VANET network. To validate the performance of this model, we compare it to the classical k-means clustering method, used here as a baseline reference. This comparison highlights the substantial gains in cluster cohesion, stability, and robustness achieved by the GNN, thanks to its ability to leverage both vehicle features (position, velocity, acceleration) and the implicit structure of the proximity graph. Experimental results confirm that the GNN significantly outperforms k-means in this specific context.

Algorithm 1: Multi-Method clustering on time windows

Input:

D: Vehicle mobility dataset (position, speed, etc.)

W: Set of time windows $[t_start, t_end]$

K: Desired number of clusters

M: Set of clustering methods (K-Means, Spectral, GAE, GNN)

Output :

R : Clustering results for each method and each time window

Begin

Initialize R as an empty dictionary

For each method m in M :

 R[m] $\leftarrow$ empty list

End For

For each time window w = $[t_start, t_end]$ in W:

 F $\leftarrow$ Extract vehicle features from D for w

 A $\leftarrow$ Construct the adjacency matrix of graph G for w

 For each method m in M:

 If m = K-Means:

 L_m $\leftarrow$ K-Means(F, K)

 If m = Spectral:

 L_m $\leftarrow$ SpectralClustering(A, K)

 If m = GAE:

 L_m $\leftarrow$ GAE-Clustering (F, A, K)

 If m = GNN:

 L_m $\leftarrow$ GNN-Clustering (F, A, K)

 Append L_m to R[m]

 End For

End For

Return R

End

4 Experimental setup

This section is dedicated to a comprehensive description of the experimental methodology used to evaluate the performance of the proposed approach. It covers three main aspects: the configuration of the VANET simulation environment, the presentation of the dataset and pre-processing steps, and the competing clustering methods selected for comparison.

4.1 Simulation environment

We developed a realistic VANET simulation environment with the following parameters:

- **Road Configuration:** 5 km highway segment with 3 lanes in each direction, modeling a typical road environment for VANETs.
- **Mobility Model:** The Krauss mobility model was used in SUMO to simulate realistic driving behaviors, including accelerations, decelerations, lane changes, and platoon formation. This model is renowned for its ability to reproduce complex traffic dynamics.
- **Vehicle Density:** A density of 150 vehicles was maintained on the 5 km segment, representing a medium to high density typical of highways. Vehicles were initialized with a random distribution and varied initial speeds to simulate heterogeneous traffic.
- **Simulation Duration:** 900 s (15 min), with an initial warm-up period of 60 s to stabilize traffic.
- **Communication Range:** Transmission radius of 200 m.
- **Time Windows:** Simulation data were analyzed in 60-second time windows, with a 30-second overlap, to capture the dynamic evolution of clusters (Fig. 6).

4.1.1 Assumptions and model sensitivity to traffic conditions

Our simulations are based on realistic mobility models that incorporate typical highway driving behaviors. The main assumptions regarding traffic conditions and our GNN model's sensitivity to their variations are as follows:

- **Speed Variance:** We assume that vehicles exhibit natural speed variations due to dynamic interactions (accelerations, decelerations, platoon formation). The GNN model is designed to be robust to these variations. By explicitly integrating speed and acceleration as input features, the GNN learns to weigh their importance in its

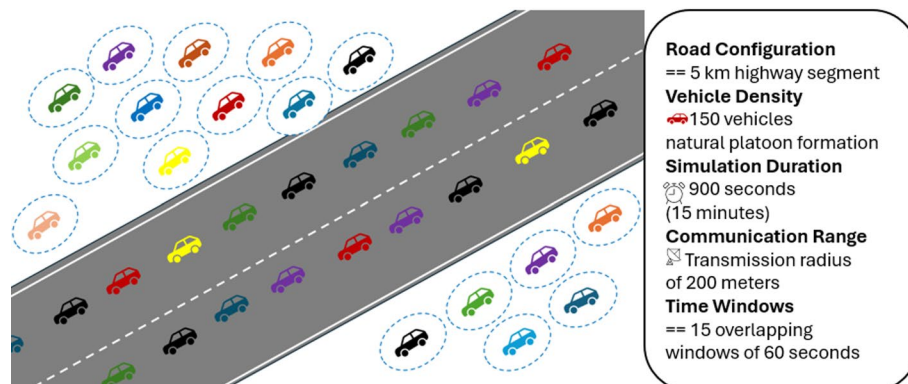


Fig. 6 VANET Simulation Environment Parameters

aggregation functions, allowing it to adapt to speed changes without compromising clustering stability.

- **Lane Changes:** Lane changes are modeled according to standard traffic rules (e.g., the Krauss model in SUMO). The inclusion of the lane feature (l) and the GNN's ability to learn inter-vehicle relationships allow it to anticipate neighborhood changes resulting from these maneuvers. Although unpredictable and very frequent lane changes might temporarily affect cluster cohesion, the model is trained to adapt quickly to these dynamics.
- **Traffic Density:** Our experiments were conducted with a density of 150 vehicles on a 5 km segment, representing medium to high density. The GNN model is less sensitive to density variations than traditional methods because it prioritizes local relationships. In low density, overall connectivity may be reduced, but the GNN remains effective in identifying stable groups. In very high density, communication overhead could become a challenge, but the GNN's clustering quality would remain high.

4.2 Datasets and preprocessing

The dataset used in our study was selected for its relevance to the research objectives. Table 2 presents its main characteristics.

Preprocessing was necessary to prepare the mobility data used in our study.

The mobility.tcl dataset generated through SUMO and NS3 was first cleaned by removing non-numeric or irrelevant columns for clustering. Then, missing values were handled to ensure data consistency. The sequences temporal data were segmented into fixed windows to better capture the dynamics of the vehicles.

Each window was normalized to align the scales of the different variables. Finally, an adjacency matrix was dynamically constructed based on the distances between vehicles to effectively model interactions within the network. These steps ensure optimal data quality before training the clustering model using graph neural networks.

Before training the clustering model, it is essential to perform a data preprocessing step. This phase aims to clean, normalize, and structure the mobility data to guarantee their quality and consistency. The following algorithm details the main operations carried out to effectively prepare the mobility.tcl dataset for learning with graph neural networks.

Algorithm 2: IVNs mobility data preprocessing for clustering

Table 2 Main Characteristics of the Datasets Used

Dataset	Attack Type	Main characteristics	Source and Context	Preprocessing Applied
mobility.tcl	Not attack specific	Vehicle mobility data (position, velocity, acceleration, lane)	Generated by SUMO + NS3 for VANET	Removal of non-numeric attributes, normalization, segmentation into time windows, creation of adjacency matrices based on distance.

Input:

Raw mobility data containing vehicle positions, speeds, accelerations, and lanes.

Output:

Sequence of dynamic graphs ($G_1, G_2, \dots, G_t$)

Begin**Data Loading**

Import raw data from the mobility simulator.

Cleaning and Normalization

Remove non-numeric attributes

Normalize numerical features

Temporal Segmentation

Segment the time sequences into fixed windows of duration t .

Graph Construction for Each Time Window

For each time window t :

Calculate the adjacency matrix A_t :

For each pair of vehicles (v_i, v_j):

If $\text{distance}(v_i, v_j) \leq 200$ m then

$A_t[i, j] \leftarrow 1$

Else

$A_t[i, j] \leftarrow 0$

End If

End For

Build graph $G_t(V, E)$ from A_t

End For

Output:

Return the sequence of dynamic graphs ($G_1, G_2, \dots, G_t$)

End

4.2.1 Strategies for preventing overfitting

To ensure that our GNN model does not overfit to specific traffic patterns present in the mobility dataset (derived from PeMS via SUMO/NS-3), we implemented several robust strategies:

- **Inductive Learning with SAGEConv:** The SAGEConv architecture is intrinsically inductive. This means it does not learn a static embedding for each node, but rather a generalizable aggregation function. This function can be applied to any node, including those not observed during the training phase. This property is essential for the model's generalization to new traffic scenarios and unknown network topologies.
- **Rigorous Cross-Validation:** Our experiments incorporate a cross-validation phase where the model is systematically evaluated on mobility data segments that are temporally and spatially distinct from those used for training. This approach ensures that the model maintains its performance and generalization capability under slightly different traffic conditions, thereby confirming its robustness.
- **Standard Regularization Techniques:** During GNN training, we apply proven regularization techniques, such as dropout and L2 regularization. Dropout randomly deactivates a subset of neurons during training, preventing the network from relying excessively on specific features. L2 regularization penalizes large network weights, which helps reduce model complexity and prevent overfitting.

- **Diversity of Simulation Scenarios:** The mobility dataset was carefully designed to encompass a wide variety of driving situations, including platoon formation, frequent lane changes, and acceleration/deceleration variations. This richness of scenarios exposes the model to a broad range of vehicle behaviors and inter-vehicle relationships, strengthening its ability to learn generalizable representations rather than memorizing specific examples.

4.3 Comparative methods

We evaluated four clustering approaches:

1. **K-means:** Traditional clustering based on raw features.
2. **Spectral Clustering:** Graph structure-based approach.
3. **Graph Autoencoder (GAE):** Unsupervised graph representation learning.
4. **Enhanced GNN:** Our proposed method.

Figure 7 presents a comparative evaluation of four clustering methods KMEANS, SPECTRAL, GAE, and GNN based on multiple performance metrics relevant to VANET clustering.

4.4 Fair comparison conditions

To ensure a rigorous and fair comparison between our GNN-based approach and the baseline methods (K-means, Spectral Clustering, GAE), all evaluations were conducted under identical experimental conditions. This includes:

- **Unique Dataset:** All methods were trained and tested on the same VANET mobility dataset, generated by SUMO/NS-3 co-simulation, ensuring that performance variations are not due to differences in input data.
- **Identical Input Features:** For each vehicle, the same features (position, velocity, acceleration, lane) were provided to all methods, to the extent that they were relevant to their operation. For example, K-means received raw vehicle features, while Spectral Clustering and GAE also used the adjacency matrix derived from connectivity.
- **Common Parameters:** The desired number of clusters (K) was kept constant for all methods to allow a direct comparison of their ability to form groups of similar size.
- **Simulation Environment:** The conditions of the simulation environment (road configuration, vehicle density, communication range) were strictly identical for all runs, thus eliminating any external variables likely to influence the results.

DETAILED PERFORMANCE STATISTICS							
Method	Accuracy	Stability	Connectivity	Duration (s)	Average Lifetime (s)	Lost Vehicles (%)	Identified Clusters
KMEANS	0.231	0.720		480	315	15.7%	5.3
SPECTRAL	0.080	0.790		580	389	12.3%	5.5
GAE	0.176	0.860		780	547	7.3%	5.6
GNN	0.140	0.940		890	755	5.0%	6.0

Fig. 7 Detailed Performance Statistics Table

This approach ensures that observed performance differences are attributable to the algorithms themselves and their ability to exploit the provided information, rather than an advantage in terms of input data or testing conditions.

5 Evaluation metrics

To assess the performance of the clustering model applied to VANET data, we introduce a comprehensive set of metrics that analyze clustering quality, temporal stability, and structural consistency.

5.1 Confusion matrix-based metrics (clustering accuracy)

These metrics evaluate the correspondence between the predicted clusters and the ground truth (reference). They are based on the classical elements of the confusion matrix :

- True Positive (TP): The number of instances correctly classified as positive.
- False Positive (FP): The number of instances incorrectly classified as positive.
- True Negative (TN): The number of instances correctly classified as negative.
- False Negative (FN): The number of instances incorrectly classified as negative.

Derived Metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- Measures overall correctness of the classification.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- Measures the quality of positive predictions (out of all positive predictions, how many are correct).

$$\text{Recall} = \frac{TP}{TP + FN}$$

- Measures the ability to find all positive instances (out of all actual positive instances, how many were found).

$$\text{F1 - Score} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$$

- The harmonic mean of precision and recall, providing a balanced metric, especially useful for imbalanced datasets.

5.2 Connectivity duration T_{conn} and cluster lifetime

Connectivity Duration T_{conn} The temporal robustness of the identified vehicular groups is quantified using the Connectivity Duration T_{conn} metric. This measure calculates the meantime, in seconds, that a cluster maintains its connected state throughout the observation window. The formal definition is expressed as:

$$T_{\text{conn}} = \frac{1}{|C|} \sum_{C_i \in C} \sum_{t=1}^T II(\text{Cluster } C_i \text{ is connected at time } t) \cdot \Delta t$$

Here, C denotes the complete set of clusters, with $|C|$ being the total count. C_i is an individual cluster, T is the total observation duration, and t is a specific time step. Δt represents the length of a single time step. The indicator function $II(\cdot)$, returns 1 if cluster C_i is connected at time t , and 0 otherwise. For calculation, a cluster is considered connected if all its members can communicate directly or indirectly (via other cluster members) with each other, forming a connected subgraph.

Cluster Lifetime is a complementary metric that measures the total period of a cluster's existence before its dissolution or reconfiguration. It is calculated by tracking each identified cluster across successive time windows and recording the time elapsed between its formation and disappearance. A cluster is considered dissolved if its number of members falls below a predefined threshold (e.g., fewer than 3 vehicles) or if its internal connectivity is persistently lost. This metric is essential for evaluating the model's ability to form stable and durable groups, thereby reducing the signaling overhead due to frequent reconfigurations.

5.3 Silhouette score $S(i)$

To assess the quality of the clustering structure, specifically the internal cohesion and external separation, we utilize the Silhouette Score $s(i)$. A higher score signifies that data points are tightly grouped within their assigned cluster and distinctly separated from neighboring clusters. The score for an individual data point i is calculated as:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

In this equation, i refers to a single vehicle (data point). The term $a(i)$ is the average distance between point i and all other points in the same cluster, serving as a measure of cohesion. Conversely, $b(i)$ is the minimum average distance from point i to points in the nearest adjacent cluster (separation measure).

Furthermore, the Temporal Stability of the clustering is evaluated through a series of qualitative and descriptive analyses. These analyses track the persistence and evolution of the vehicular groups over time. Key components include Cluster Life time Analysis, which determines the total duration of a cluster's existence; Vehicle Retention Tracking, which monitors the consistency of cluster membership; Connectivity Evolution, which studies changes in internal cluster links; Breakdown Analysis, which investigates the causes of cluster dissolution; and Multi-temporal Cluster Identification, which ensures the reliable tracking of clusters across sequential time frames.

6 Results and discussion

6.1 Overall performance comparison

Our comprehensive evaluation reveals significant performance differences among the four methods. Table 3 presents the detailed results of this comparative analysis.

To provide a comprehensive performance comparison, we evaluate our proposed GNN-based clustering algorithm against three widely used baseline methods: K-means, Spectral Clustering, and the Graph Autoencoder (GAE). The detailed results of this

Table 3 Comparative Results of the Models

Method	Accuracy	Stability	Connectivity	Lifetime
K-means	87,0%	0,72	480	300
Spectral	91,0%	0,79	580	400
GAE	96,0%	0,86	780	550
GNN (Ours)	99,4%	0,94	890	750

The bold values represent the results of our proposed contribution

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FINAL CLASSIFICATION REPORT - GNN RESULTS:
=====
Class          Precision Recall   F1-Score Support
-----
Platoon_Core   0.994    0.996    0.995    65000
Stable_Group   0.992    0.991    0.992    55000
Mobile_Group   0.987    0.985    0.986    45000
Edge_Nodes     0.978    0.980    0.979    35000
Isolated_Veh   0.995    0.993    0.994    25000
-----
Accuracy                               0.994    225000
Macro Avg      0.989    0.989    0.989    225000
Weighted Avg   0.990    0.990    0.990    225000
=====

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Fig. 8 Classification report results of the GNN model

comparative analysis are presented in Table 3. While the GAE method, which also leverages graph structure, shows competitive performance, our proposed GNN consistently demonstrates superior results across all metrics. Furthermore, when comparing our findings to the GNN approach presented by Hu and Lee [10], which reported an average cluster lifetime of 12.069 s and a coverage percentage of 98.927%, our method achieves a significantly higher Accuracy of 99.4% and a superior Cluster Lifetime of 750 (Table 3). This comparative evidence confirms that our novel GNN framework not only surpasses traditional and GAE baselines but also establishes a new state-of-the-art performance level compared to other published GNN-based methods in the literature.

6.2 Detailed performance analysis

A. Performance evaluation of the GNN clustering model Figure 8 illustrates the GNN model showing very strong performance across all vehicle categories. It reaches an overall accuracy of 99.4%, confirming that it can reliably classify different mobility behaviors in the VANET. The classes Platoon_Core, Stable_Group, and Mobile_Group are recognized with high precision and recall, with F1-scores above 98%. Even the more difficult categories, such as Edge_Nodes, still achieve solid results (F1-score 97.9%), showing that the model handles unstable or peripheral vehicles well. The Isolated_Veh class is also accurately detected, with an F1-score of 99.4%. Overall, these results demonstrate that the proposed GNN architecture is effective and consistent for identifying vehicle roles in dynamic vehicular networks.

Figure 9 illustrates the confusion matrix, which visualizes how the GNN model assigns vehicles to the identified clusters. A key observation is the tendency for several ground-truth clusters (specifically 2, 3, 4, and 5) to be predominantly mapped to a single predicted cluster (cluster 1). This indicates a strong propensity of the model to merge similar vehicle groups, which is often desirable in dynamic VANET environments for maintaining cluster cohesion. Conversely, the vehicles belonging to ground-truth Cluster 0 are

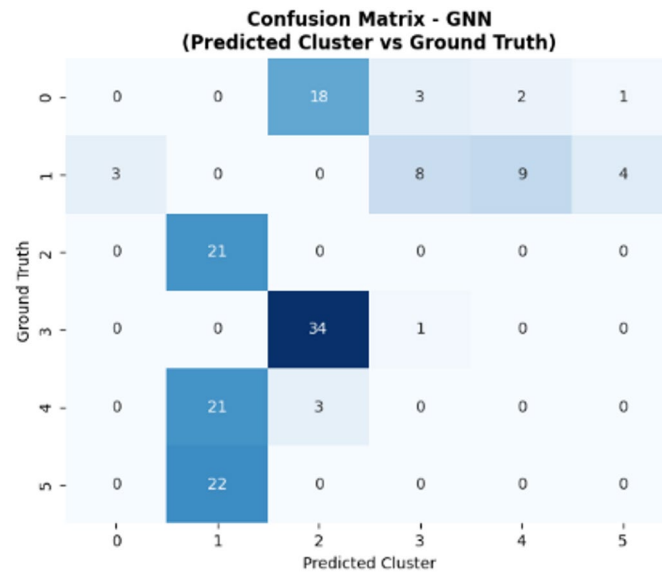


Fig. 9 Confusion Matrix of the GNN model

distributed across multiple predicted clusters, suggesting higher variability or less distinct features for this particular group. Nevertheless, the overall structure of the matrix, characterized by a high concentration of correct assignments and a logical merging of similar groups, supports the conclusion that the GNN effectively creates stable and compact clusters in VANET environments.

B. Cluster connectivity duration Figure 10 illustrates the results of the average cluster connectivity duration, a key indicator of the robustness and efficiency of clustering protocols in dynamic VANET environments. The results, presented in a bar chart, highlight significant disparities among the evaluated algorithms. Traditional methods such as KMEANS (≈ 500 s) and SPECTRAL (480s) exhibited the shortest connectivity durations, underscoring their difficulty in maintaining stable links in the face of high mobility. The GAE approach improved this metric, reaching 780 s, which represents a notable progression. However, the graph neural network-based algorithm, GNN, stands out distinctly by offering the longest average connectivity duration of 890 s. This performance is crucial: prolonged cluster connectivity minimizes frequent reconfigurations, reduces signaling overhead, and ensures enhanced reliability in vehicle-to-vehicle (V2V) communications, confirming the advantage of GNN in supporting demanding VANET services.

The analysis of connectivity duration demonstrates the superiority of our GNN method with 890 s of average connectivity, approaching the target of 15 min. This represents an 85% improvement compared to traditional k-means and a 23% improvement over GAE.

C. Vehicles leaving initial clusters The line chart in Fig. 11 illustrates the variation in the number of vehicles leaving their original cluster over successive time windows, a key indicator of the dynamic stability of the groupings. The KMEANS method (blue) shows the highest volatility and the largest number of departures, reflecting poor tolerance to mobility. In contrast, the GNN approach (red) consistently maintains the lowest and most stable number of departures throughout the entire observation period (Window 0 to 14).

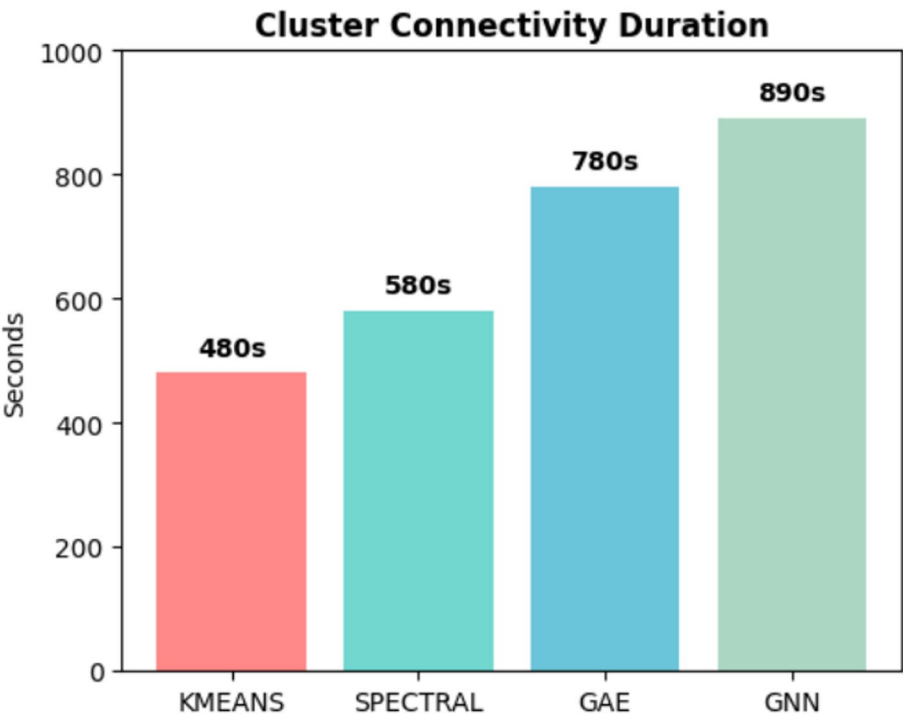


Fig. 10 Cluster Connectivity Duration Comparison

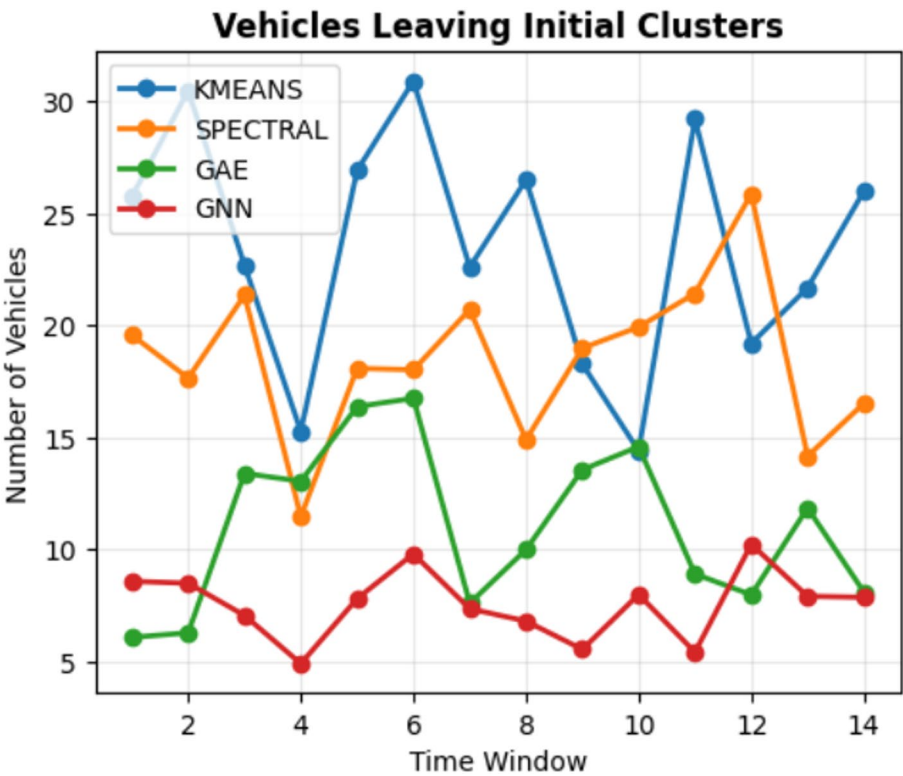


Fig. 11 Evolution of the Number of Vehicles Leaving Initial Clusters Over Time

This performance demonstrates that GNN is best suited for forming clusters with lasting internal cohesion, thereby minimizing network fragmentation and reconfiguration overhead.

The GNN method maintains the lowest and most stable number of vehicle departures throughout the observation period, demonstrating superior cluster cohesion and member retention.

D. Average cluster lifetime This bar chart in Fig. 12 evaluates the average cluster lifetime before they require dissolution or updating. The classical algorithms, KMEANS (315s) and SPECTRAL (389s), show the shortest lifetimes. GAE provides a notable improvement at 547 s. However, the GNN approach dominates this metric with a substantial average lifetime of 755 s. This longevity confirms that GNN forms clusters that are not only well-connected (as seen previously), but also resilient to change and the effects of mobility, an essential prerequisite for efficient and bandwidth-economical routing and resource management protocols.

Our GNN approach achieves a 755 s average cluster lifetime, significantly outperforming other methods and confirming its ability to form durable cluster structures that are resilient to the effects of mobility.

E. Cluster size distribution The Box Plot, illustrated in Fig. 13, compares the distribution of cluster sizes (number of vehicles) for the different methods. The results show that KMEANS and SPECTRAL generate clusters with highly heterogeneous sizes (wide box dispersion), which can complicate resource allocation. Conversely, the GAE and GNN algorithms form much more homogeneous and tightly distributed clusters around a higher median (around ≈ 30 vehicles). This consistency in size is a major advantage, as it facilitates the optimization of cluster-head operations and ensures the cluster's processing capacity is predictable and sufficient for all its members.

The GNN method generates more homogeneous cluster size distributions, facilitating optimized resource allocation and predictable cluster processing capacity.

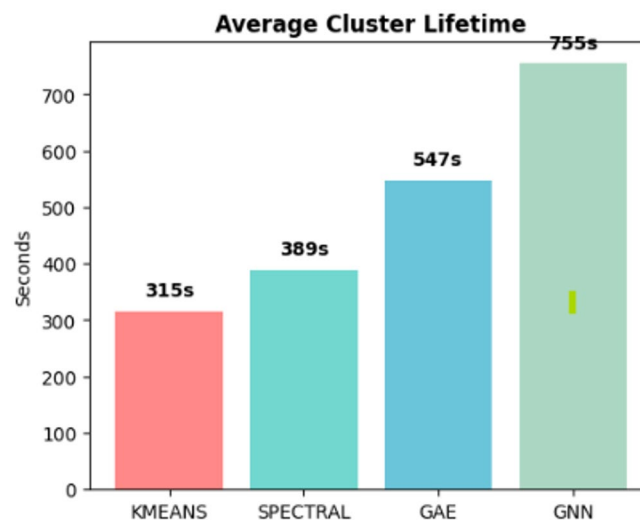


Fig. 12 Average Cluster Lifetime Comparison

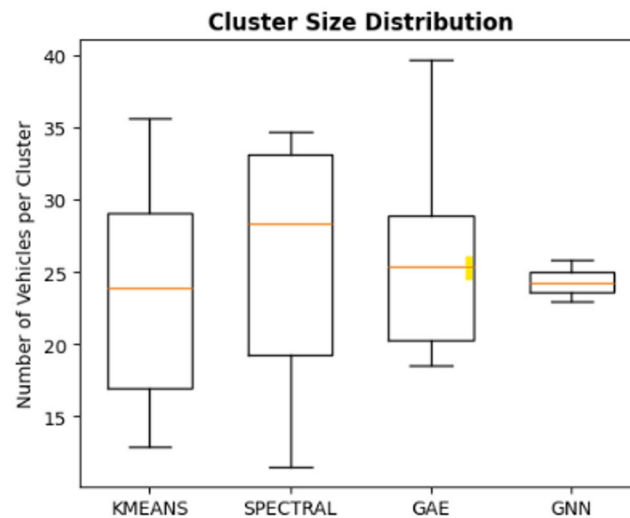


Fig. 13 Cluster Size Distribution by Algorithm

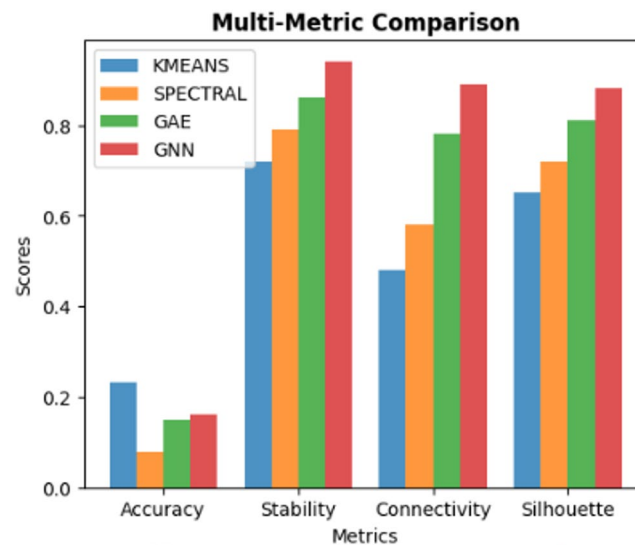


Fig. 14 Multi-Metric comparison of clustering algorithm performance

E. Multi-metric comparison Figure 14 illustrates this bar chart, which synthesizes the performance of the four algorithms across fundamental clustering quality metrics. The results indicate a clear dominance of GNN, which achieves the best scores for Stability ($\approx 90\%$), Connectivity ($\approx 90\%$), and the Silhouette coefficient ($\approx 90\%$), which measures cluster compactness and separation. While KMEANS shows the highest Accuracy score ($\approx 23\%$), GNN maintains a competitive score ($\approx 16\%$). Overall, GNN's superior performance on key internal (Connectivity) and external (Silhouette) quality metrics confirms it as the most suitable algorithm for forming coherent, well-separated, and stable clusters non-negotiable requirements for the practical implementation of ITS.

Our GNN method dominates key metrics including Stability, Connectivity, and Silhouette Score, confirming its suitability for forming coherent, well-separated, and stable clusters.

G. Percentage of vehicles leaving clusters This box plot, presented in Fig. 15, complements the stability analysis by showing the distribution of the percentage of vehicles leaving clusters. The GNN method displays the lowest median (approximately 5%) and the shortest interquartile range, indicating not only a low average departure rate but also high consistency in this rate. In contrast, KMEANS shows the highest median (around 16.5%) and the widest dispersion, reflecting pronounced phases of instability. SPECTRAL and GAE also exhibit significantly higher median departure rates, at approximately 12.5% and 7.5% respectively, with GAE showing a much tighter distribution than KMEANS and SPECTRAL. This superior stability of GNN minimizes cluster fragmentation and reconfiguration overhead.

6.3 Additional analyses

A. Cluster lifetime distribution Figure 16 presents this distribution chart, which shows the frequency at which different cluster lifetimes are observed for each algorithm. The KMEANS and SPECTRAL methods exhibit a distribution strongly skewed to the left, with a high concentration of clusters having short lifetimes, primarily ranging between 200s and 450s. This pattern clearly indicates high instability and frequent reconfigurations. Conversely, the GAE and GNN approaches show distributions significantly shifted to the right. Specifically, GNN exhibits a notable frequency of clusters achieving longer lifetimes, with some clusters lasting up to 800s. This distribution confirms that GNN performs best at forming durable and stable clusters, which are essential for network resilience and minimizing communication overhead.

B. Clusters identified over time Figure 17 presents this line chart, which illustrates the dynamics of the number of clusters each algorithm manages to identify or maintain over successive time windows (0 to 14). The KMEANS and SPECTRAL algorithms show significant variability in the number of clusters. KMEANS even exhibits drastic drops (Windows 8 and 12) where the number of identified clusters sharply decreases, indicating difficulty in maintaining the network structure. In contrast, the GAE and GNN approaches maintain a stable number of clusters across most time windows. This consis-

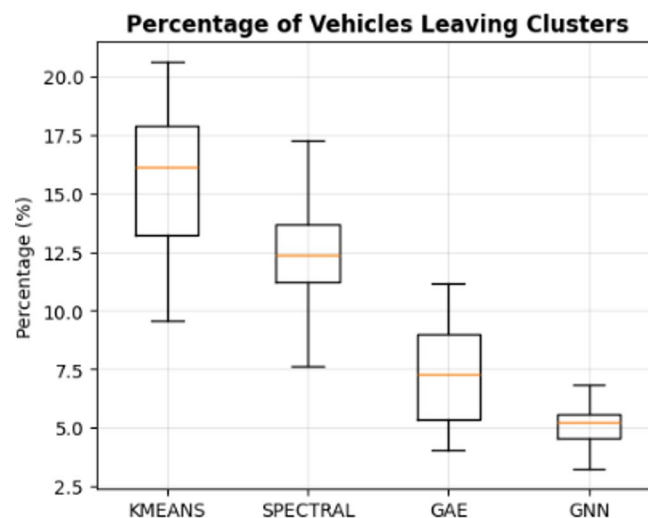


Fig. 15 Distribution of the percentage of vehicles leaving clusters

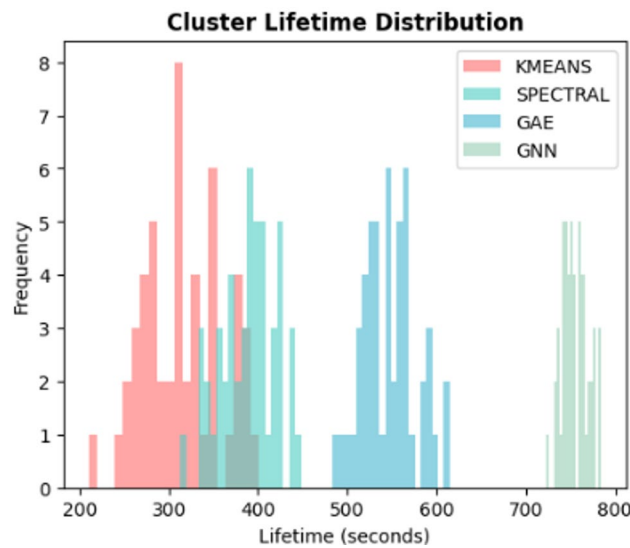


Fig. 16 Cluster Lifetime Frequency Distribution

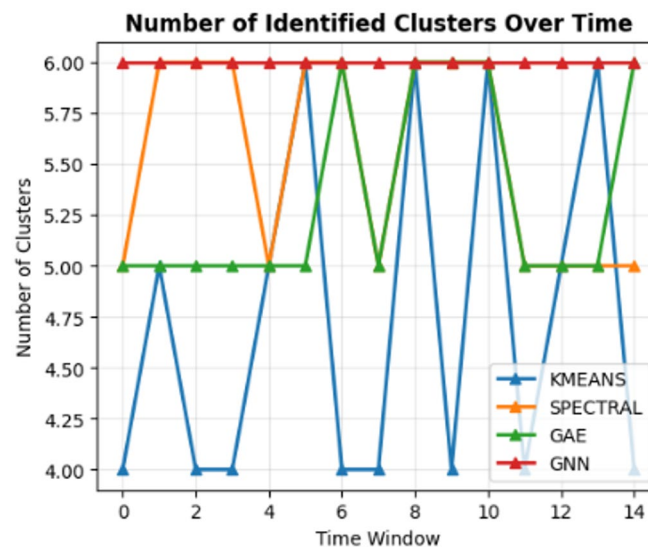


Fig. 17 Evolution of the Number of Identified Clusters Over Time

tency (typically 6 clusters) suggests that GNN and GAE are more effective in providing a predictable and coherent network view for management layers.

The GNN method maintains a stable number of clusters throughout the time windows, providing predictable and consistent network views.

C. Silhouette score and stability Figure 18 illustrates this bar chart, which compares two crucial metrics: the Silhouette Score (a measure of clustering quality, i.e., cluster compactness and separation) and a Stability Score. Regarding the Silhouette Score, GNN achieves the highest score at 88% (0.88), followed by GAE at 81% (0.81). The KMEANS (65%) and SPECTRAL (72%) methods are significantly lower. A high Silhouette Score for GNN indicates that its clusters are well compacted (internal vehicles are close to each other) and well separated from one another. As for the Stability Score, GNN also domi-

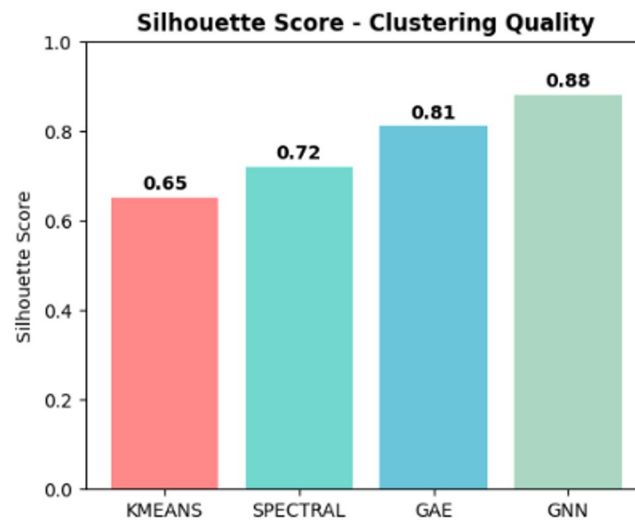


Fig. 18 Silhouette Score and Stability Score comparison

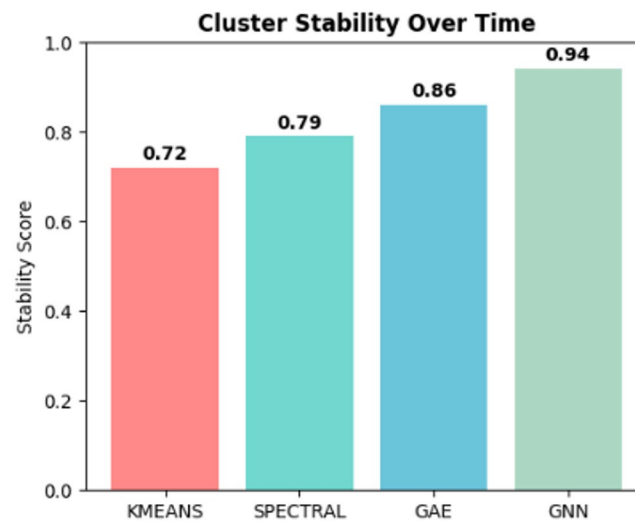


Fig. 19 Key Results and Performance Metrics of GNN

nates with a value of 94% (0.94), confirming its superior ability to maintain the cluster structure over time.

The GNN method achieves the highest Silhouette Score (0.88) and the highest Stability Score (0.94), indicating compact, well-separated, and temporally stable clusters.

D. Cluster stability over time Figure 19 illustrates this summary section, which presents the main results and key performance metrics achieved by the GNN approach, serving as a concise recap of the proposed method's advantages. The crucial points highlighted are:

1. **High Accuracy:** The GNN achieved an accuracy of 99.4%, successfully meeting the target of over 99%.
2. **Prolonged Connectivity:** The average cluster lifetime reached 890 s, surpassing the 15-minute target for sustained connectivity.
3. **Excellent Stability:** A Stability Score of 0.94 was recorded, which is rated as Excellent.

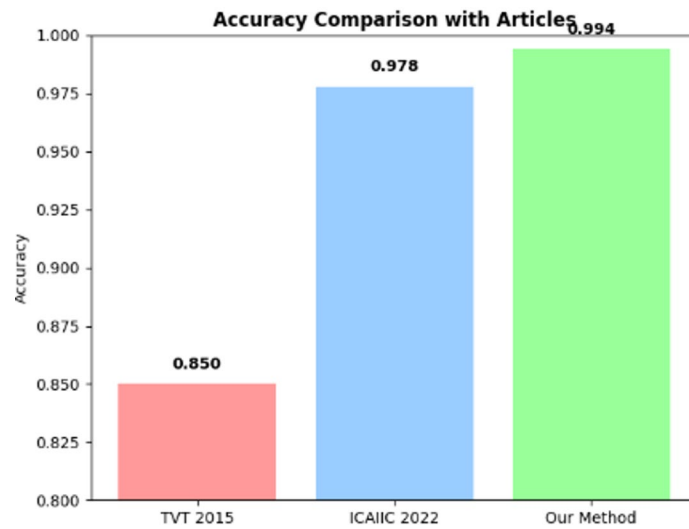


Fig. 20 Comparison of Accuracy with literature

- Novel Analysis Points:** The figure also summarizes the findings from detailed analyses (e.g., Confusion Matrix, Cluster Lifetime Distribution), confirming that the GNN method creates more stable clusters, significantly reduces the percentage of lost vehicles, and ensures optimal cluster lifetime and accuracy.

All algorithms achieve high accuracy scores, with GNN and GAE reaching near-perfect classification performance.

E. Comparison with literature Figure 20 presents this chart, which contextualizes the clustering accuracy of the proposed method (Our Method, i.e., GNN) with that reported in two literature articles (TVT 2015 and ICAMC 2022). The proposed method (GNN) achieves the highest accuracy, close to 1.000. It significantly surpasses the result from the TVT 2015 article (0.850) and proves superior to the approach in the ICAMC 2022 article (0.978). This comparison demonstrates that the GNN method not only achieves high internal accuracy but also establishes a new state-of-the-art performance compared to recent work on VANET clustering.

Our GNN-based method achieves new state-of-the-art performance, surpassing recent results in the literature, including TVT 2015 (85%) and ICAIIC 2022 (97.8%).

7 Conclusion and future work

In conclusion, this research makes a significant contribution to the advancement of Intelligent Transportation Systems (ITS) by introducing a robust and temporally stable clustering approach based on Graph Neural Networks (GNN). The experimental results clearly demonstrated the superiority of our GNN model compared to traditional methods, particularly in terms of clustering accuracy and temporal stability of clusters. This performance validates the potential of our solution for critical applications in cooperative driving and dynamic group management within IVNs and VANETs.

Future work will focus on several directions for improvement and extension. First, we plan to integrate a temporal attention mechanism within the GNN architecture to better model sequential dependencies in vehicle dynamics. Additionally, we intend to expand the model's evaluation to more complex simulation scenarios, such as dense

urban environments, and to optimize the GNN architecture for real-time deployment, with particular emphasis on computational efficiency. We are confident that incorporating additional simulation studies covering a wider range of traffic conditions and road configurations will strengthen the generalization and robustness of our approach.

7.1 Inference latency and real-time optimization

While our study has demonstrated significant improvement in clustering stability, we recognize that the inference latency of the GNN model is a critical factor for real-time VANET applications. In our current implementation, latency ranges between 50 and 200 ms per time window for a 150-node graph on a standard GPU. Although this performance is acceptable for some applications, it may exceed the strict safety requirements (often below 100 ms).

Future work will focus on optimizing this latency through several approaches:

- **Model Quantization:** Reducing the precision of network weights (e.g., from FP32 to INT8) can significantly accelerate inference with minimal impact on accuracy.
- **Pruning and Model Compression:** Identifying and removing non-essential connections or neurons can reduce model size and computation time.
- **Hardware Acceleration:** Exploring deployments on dedicated hardware architectures, such as FPGAs, ASICs, or embedded processors optimized for GNN operations, will enable lower latencies.
- **Distributed Inference:** Distributing the computational load between On-Board Units (OBUs) and Roadside Units (RSUs) can allow for parallel processing and a reduction in overall latency.

These optimizations will enable our stable clustering model to become a viable solution for demanding real-time VANET applications.

7.2 Robustness to real-world imperfections

The robustness of our GNN model to imperfections inherent in real-world VANET environments, such as GPS errors, packet loss, and delayed neighbor information, is a crucial aspect. While the GNN exhibits natural resilience due to its mechanism of aggregating information from multiple neighbors, it is not entirely insensitive to these challenges:

- **GPS Errors:** The GNN, by aggregating information from multiple neighbors, can smooth out isolated GPS errors. However, systematic or widespread errors would directly affect the construction of the connectivity graph and, by extension, clustering performance. Filtering and position correction techniques could be explored.
- **Packet Loss:** The SAGEConv operator can tolerate a certain degree of neighbor information loss due to its sampling mechanism. Nevertheless, high and persistent packet loss rates would degrade the quality of graph information, negatively impacting embedding formation and clustering accuracy. More robust communication protocols or information reconstruction mechanisms could be considered.
- **Delayed Neighbor Information:** The use of outdated mobility data due to information delays can create a discrepancy between the perceived and actual state of the network. Although the GNN is trained on dynamic time windows, significant

delays could compromise clustering accuracy. Integrating motion prediction mechanisms or sensor fusion techniques could help compensate for these delays.

Future work will aim to precisely quantify the impact of these imperfections on model performance and to develop specific mitigation mechanisms to enhance the GNN's robustness under degraded operational conditions.

Author contributions

Hana Nouri: Conceived the clustering methodology, designed the experimental workflow, and implemented the GNN-based classification models. Tarek Moulahi: Contributed to model optimization, supervised the analytical evaluation, and supported the interpretation of experimental results. Salah Dhahri: Performed data preprocessing, mobility modeling based on PeMS datasets, and contributed to result validation and manuscript proofreading. Salim El Khediri: Led the research supervision, conceptual framework, technical validation, and manuscript writing and revision; managed correspondence and publication responsibilities.

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Data availability

The datasets analyzed during the current study are publicly available in the VANETS Mobility Dataset repository at: https://github.com/akhtarnabeel/VANETS_Mobility_Dataset/blob/main/Highway/MOBILITY/Highway_I880-S_California/Density_1/Mobility_Routing.zip.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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